

Tutorial 4

1 Discounted Utility

Suppose utility is $U(x_0, x_1) = u(x_0) + \delta u(x_1)$ where $u(x) = \ln(x)$. Recall that $u'(x) = \frac{1}{x}$. Suppose the consumer buys consumption to maximize utility subject to a budget constraint $p_0x_0 + p_1x_1 = m$. Thus, when the consumer maximizes utility his demand will satisfy $\frac{u'(x_0)}{\delta u'(x_1)} = \frac{p_0}{p_1}$.

1. Find the demand function: $x(p, m) = (x_0(p, m), x_1(p, m))$. So, $x_k(p, m)$ should tell you how much the consumer demands of good k when price is p and expenditure level is m .
2. What happens to the demand for good 0 as δ increases?
3. Suppose prices take the form $p = (1, \frac{1}{1+r})$. What happens to the demand for good 0 as r increases (equivalently what happens to the demand for good 0 when p_1 decreases)?
4. Suppose the consumer has income $(y_0, y_1) = (0, 100)$ and the interest rate is r . What is the present value of this income?
5. Continue to assume the consumer has income $(0, 100)$. What happens to the consumer's demand for good 0 as r increases (note that increasing r changes the present value of the income stream)?
6. Assume the consumer has income $(0, 24)$. Suppose that $\delta = \frac{1}{2}$ and that the interest rate is $r = 1$. How much will be demanded of good 0 and 1?

2 Interest rates

Suppose you can borrow at interest rate r_b and save at rate r_s . You are endowed with $y = (1, 1)$

1. Draw the budget line when $2 = r_b > r_s = 1$ (note that the budget line is not in-fact a straight line).

2. Draw the budget line when $2 = r'_s > r'_b = 1$.
3. Suppose the consumer has $U(x_0, x_1) = x_0 + x_1$ (note discount factor $\delta = 1$). Does the consumer prefer interest rates (r_b, r_s) or (r'_b, r'_s) ? That is, under which pair of interest rates can the consumer buy a bundle with greater utility?

3 Comparing assets

Consider three assets: $y^A = (1, 0, 0)$, $y^B = (0, 3, 0)$, and $y^C = (0, 0, 4)$.

1. For what interest rates r is A the best? What about B and C ? To answer consider the present values.
2. Answer the same question when you are endowed with $(2, 0, 0)$. That is, if you take asset y^A then you get $(2, 0, 0) + y^A$ and so forth.
3. Now, assume that you have no money $(0, 0, 0)$ and asset B costs $\frac{4}{1+r}$. Can you find an arbitrage opportunity?
4. Let p_A and p_C be the prices of asset A and C in period 0. What happens to the ratio $\frac{p_A}{p_C}$ as r increases (assume p_A and p_C are determined by the no-arbitrage condition)?